Machine learning for integrating clinical and biological data

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Association Française de Cytométrie

L'AFC EVÉNEMENTS WEBINAIRE BOURSES GROUPES DE TRAVAIL ANNONCES FORMATIONS EN CYTOMETRIE PARTENAIRES ISAC ESPACE ADHÉRENT



R4CYTO

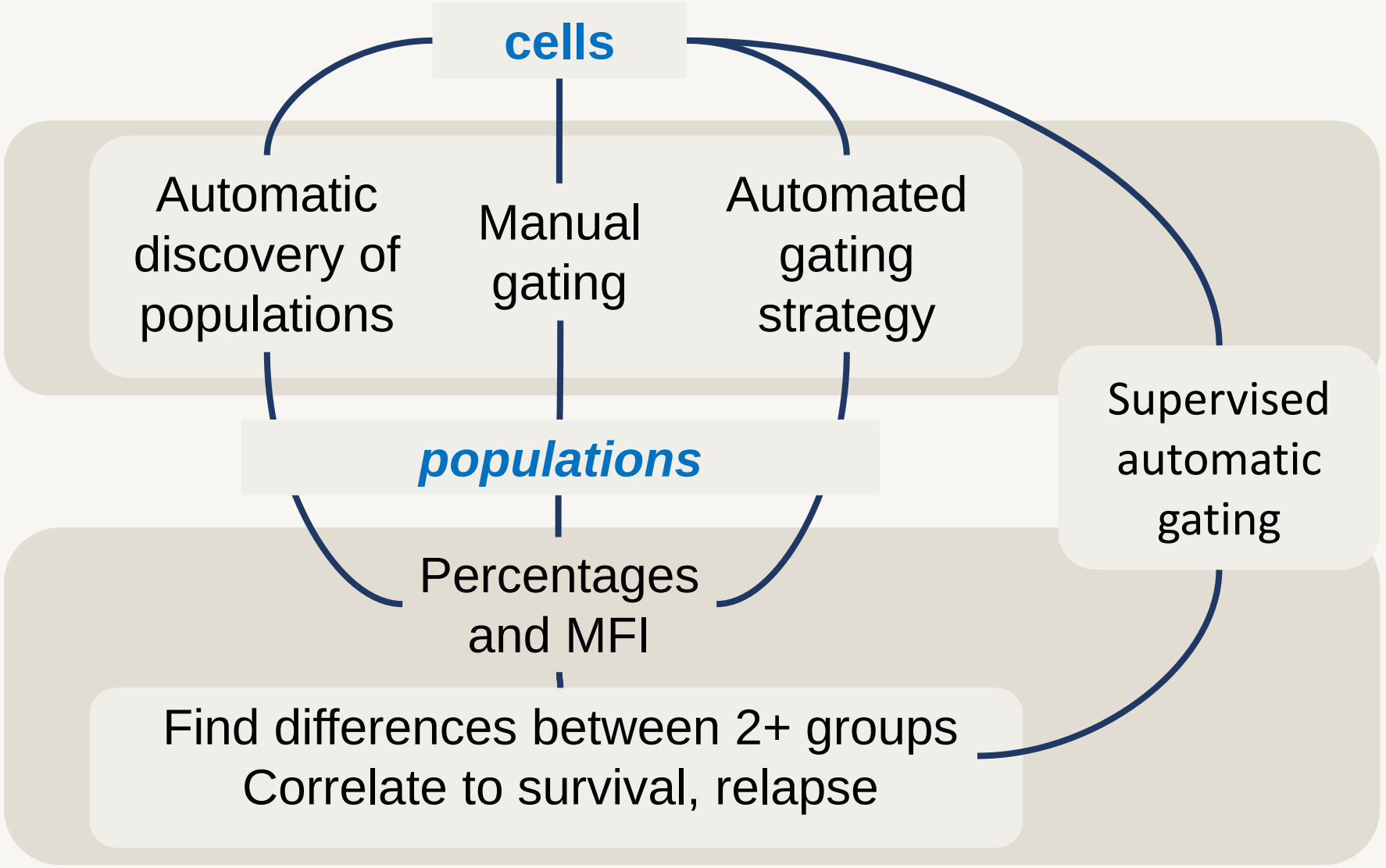
Vous travaillez en cytométrie, êtes adhérent.e de l'AFC et vous souhaitez apprendre à lire et utiliser des scripts R ? La formation R4cyto vous apprendra les bases du langage R et l'utilisation de packages spécifiques appliqués à la cytométrie.

Formation en distanciel par petits groupe. Inscrivez-vous dès maintenant !

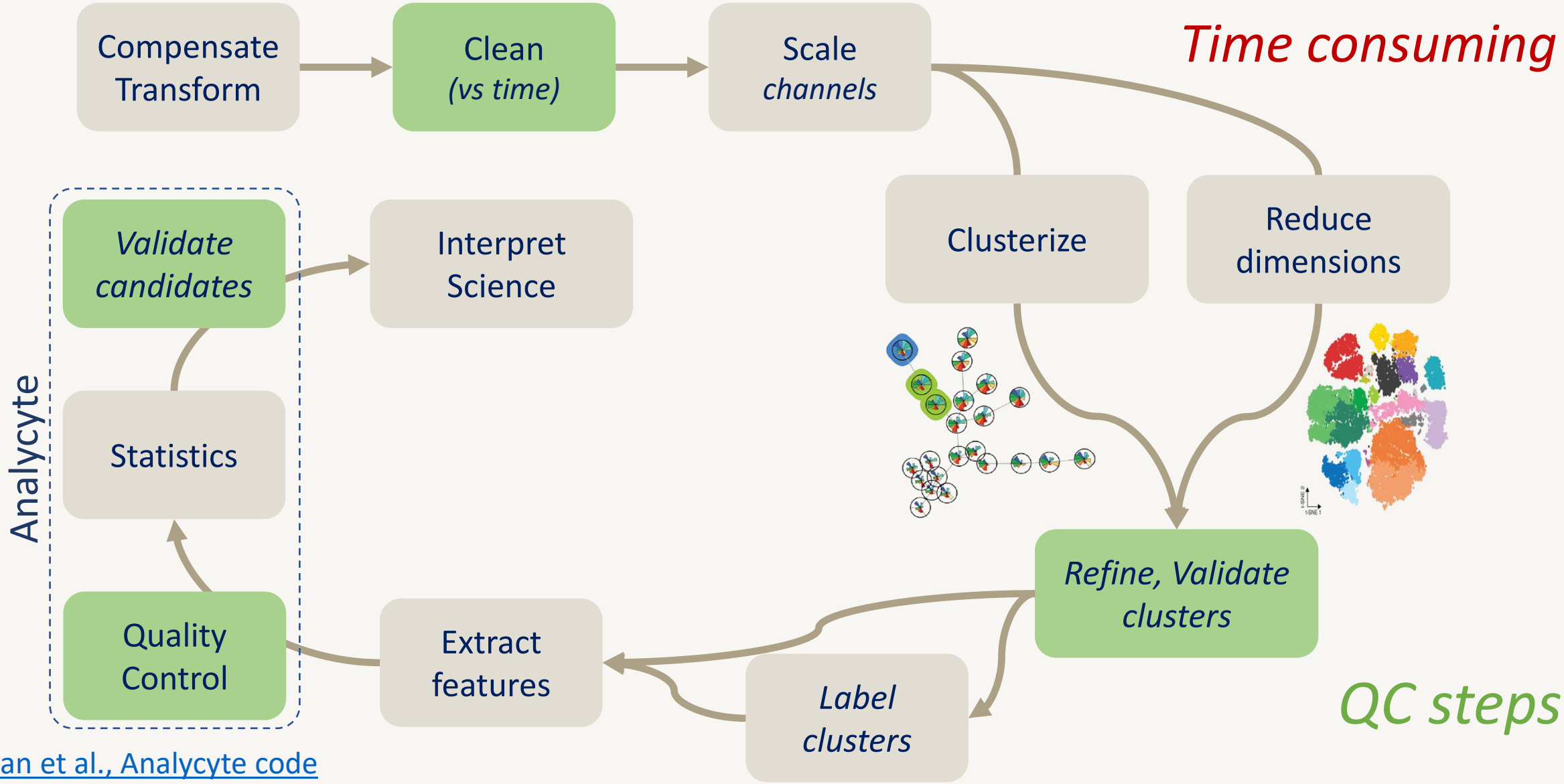
Typical workflows

cells clustering

features analysis



Unsupervised analysis workflow



[Lohman et al., Analycyte code]

Important points

- Experiment = answer a question
  - the experimental design is important
  - get a reproducible and generalized result
  - especially if technical steps are performed at different time points
  - especially if we are asked for a validation experiment (anticipate batch effect)

- Annotate acquisition files
  - the sooner, the better (S. van Gassen)
  - help identifying problems, outliers
  - correlated with the labelling dates, a power shutdown or at random

Important points

- Quality control = data is homogeneous/in the range
  - what range? range of the current experiment
  - range of a previous experiment?
  - find potentially attended perturbations but also unattended ones
  - QC is important ASAP

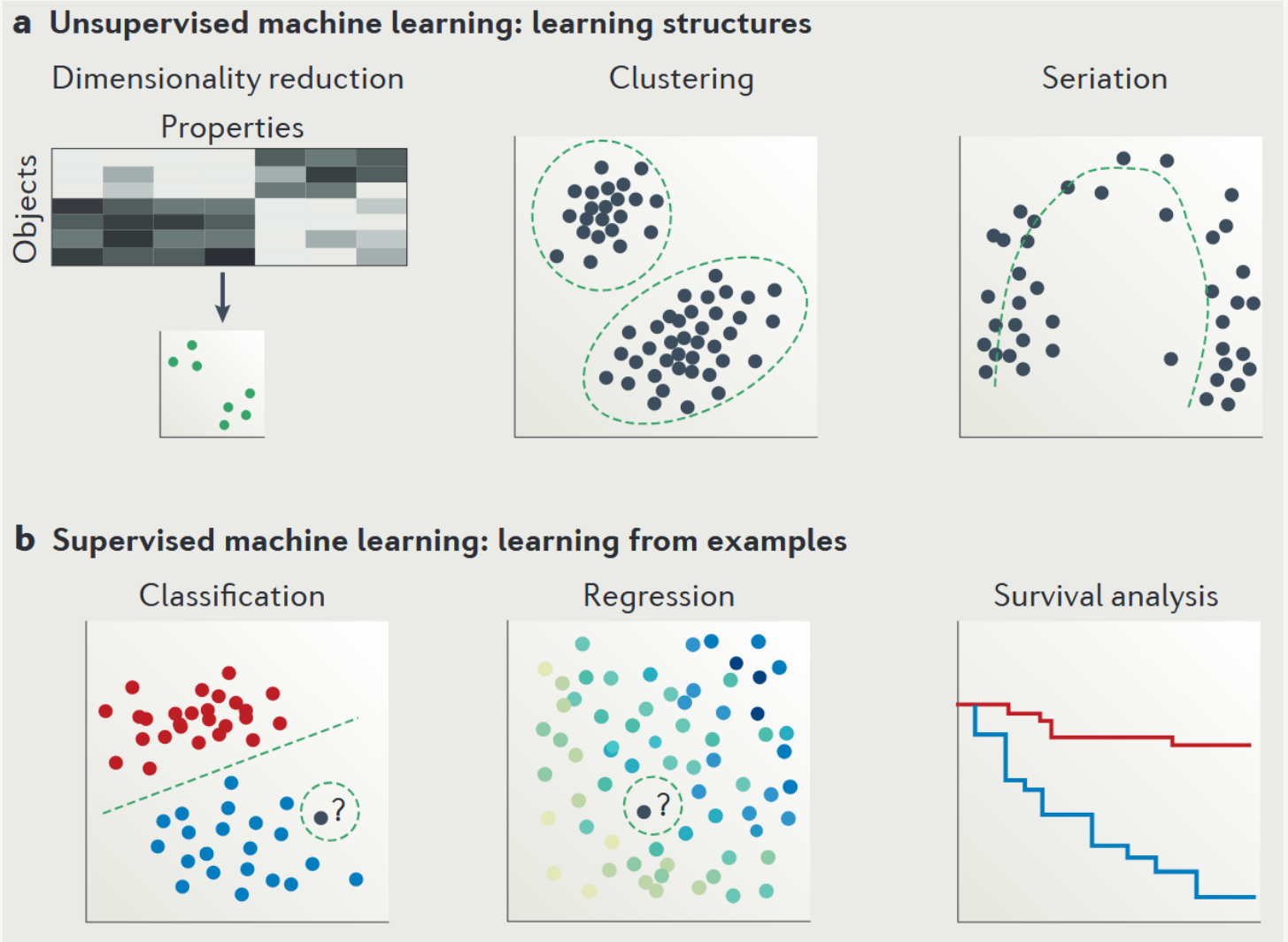
- Design should anticipate perturbations
  - labelling dates
  - acquisition runs
  - etc.

Using computational flow cytometry tools

- Generate the data
  - Consult a statistician or bioinformatician to help in experimental design
- Clean the data
  - Consider pre-gating on a cell population of interest
- Visualize the data
  - Try different techniques (tSNE, UMAP, PacMap, etc.), random seeds
- Analyze the data
  - Choose the type of analysis to be performed: biomarker discovery, predictive modelling, etc.
  - Adapt parameters of the algorithm where necessary

Saeys et al., 2016

What is Machine Learning?



Unrestricted discovery of structures

Discovery of structure constrained to predict the response

[Saeys et al., 2016]

Benefits, Limitations

- Automates Complex Rule-Making: Learns patterns directly from data instead of hand-coded logic
- Captures Non-Linear Relationships: Uncovers complex correlations in high-dimensional and unstructured data.
- Adaptability: Continuously improves and updates as new data becomes available.
- Scalability: Processes massive volumes of data far beyond human speed or manual capacity.

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- Adaptability: Continuously improves and updates as new data becomes available.
- Scalability: Processes massive volumes of data far beyond human speed or manual capacity.
- Bounded by Input Data: Models only know what the training data reflects (susceptible to bias and noise)
- Data Demand: Requires large, representative datasets to generalize effectively.
- Curse of Dimensionality: Performance degrades and sample requirements explode as features multiply.
- Data Leakage Risk: Vulnerable to train/test contamination leading to false high confidence.

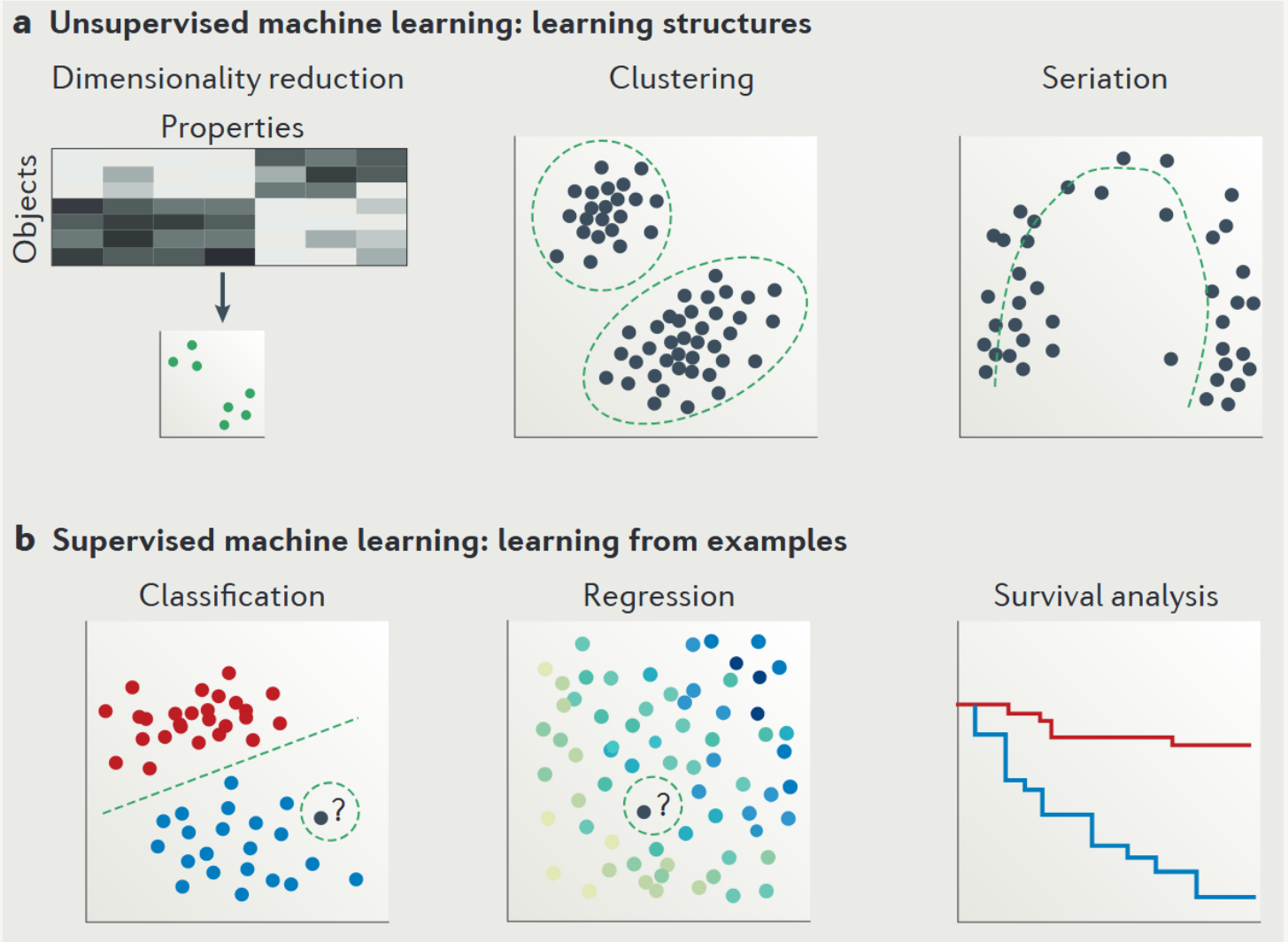
What is Machine Learning?

- Algorithm process input data according a recipe and parameters and returns output data

- The recipe was designed by a **human** that had some **goals** and **constraints** in mind

- ML means learning from the data: algorithm is naive
  - knows only what we bring to it
    => **representative data**, group balanced data
    => numerical experiment, think about its design
  - needs reasonable values for its parameters

ML parameters?



from experience or wisdom of crowd

adequacy to external biological knowledge

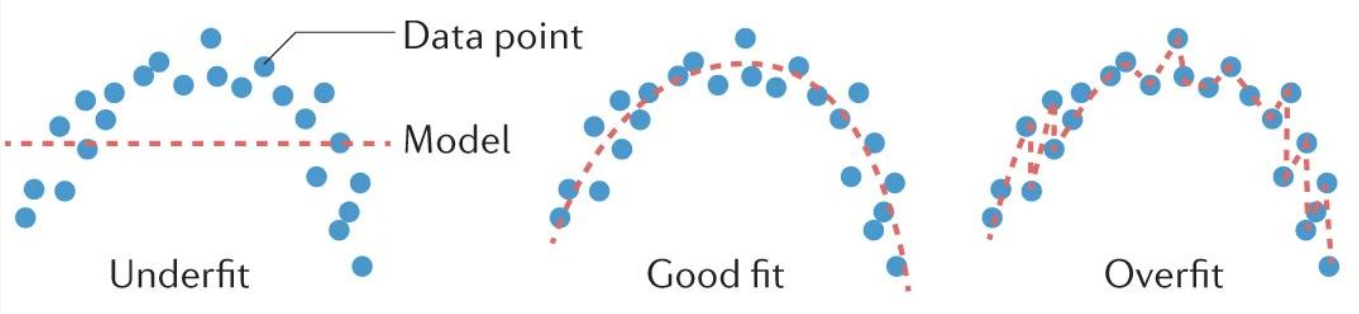
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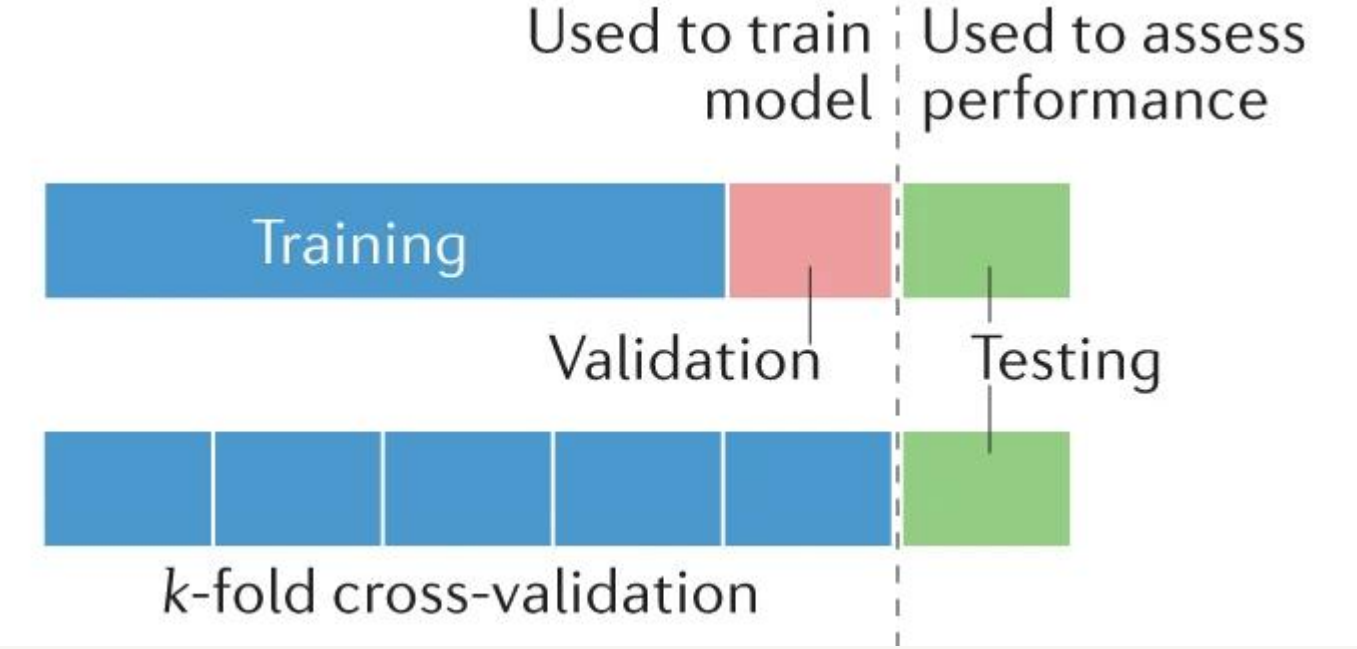
OR

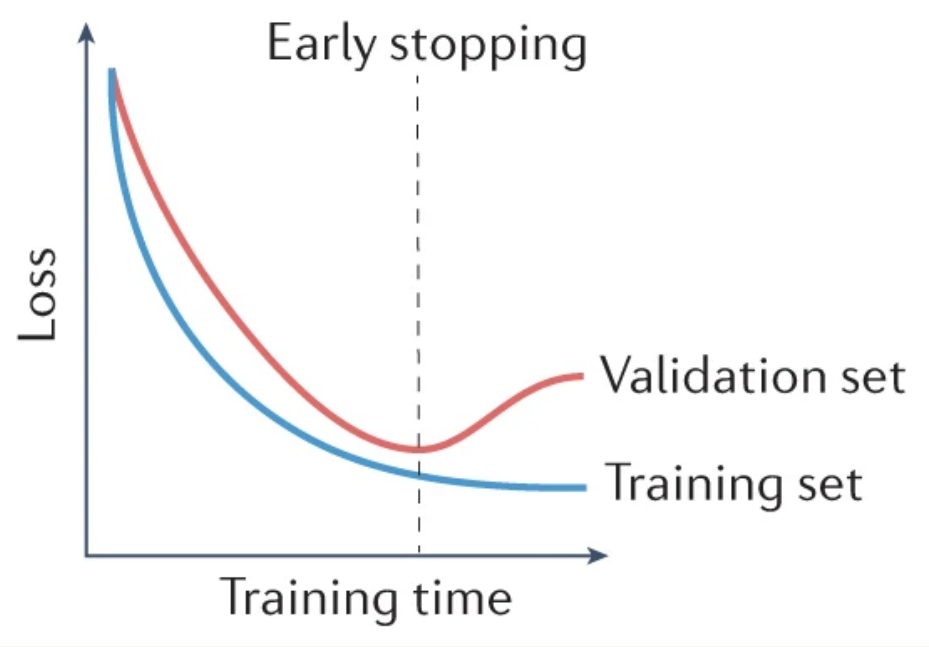
from data itself

[Saeys et al., 2016]

Cross-validation







[Greener et al., 2020]

XAI - Feature Selection & Importance

- Models learn structures directly from data without assuming linear relationships or normal distributions, capturing complex feature interactions and non-linear effects.
- Automated Feature Selection: Identifies top predictive features automatically, helping condense high-dimensional data into a sparse, minimalist set of key variables.

- Never rely solely on raw default importance.
- Use SHAP / Permutation Importance:
  Ensures robust, model-agnostic ranking of key features.
- Remove or group redundant features first, to keep the model minimalist and interpretable.

ML in questions

- Should I use ML?
  - I am dummy if I don't try it
- What should I learn?
  - Greener et al., Nature Rev. Mol. Cell Bio, 2021
  - Liu et al., JAMA, 2019
  - Articles of the brilliant speakers here
  - Nature Methods Point of Significance series
- Should I do it on my own, using AI?

Stat. Test. vs Mach. Learning: goal matters

- Statistics (inference): to understand and quantify the effect of each variable. We focus on coefficients, confidence intervals, and p-values. Assumptions (independence, linearity, etc.) are very important.
- Machine learning (prediction): predicting new observations as accurately as possible. We focus on the prediction error on unseen data, measured on a test set or by cross-validation (danger). Assumptions matter less; performance matters more.

- Interpret coefficients using p-values and confidence intervals = inferential statistics.
- Seeking the best performance on unseen data, with hyperparameter tuning via cross-validation = machine learning.

Bzdok et al., 2018

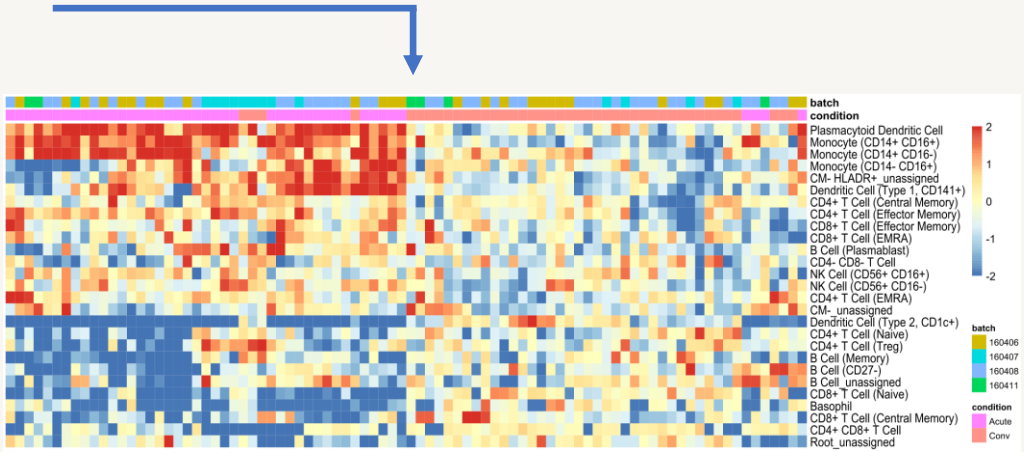
Recap of Machine Learning

- Recipe to achieve a goal
- Parameters
  - a priori from experience
  - learned from data using cross-validation
- Input data
  - for each sample/FCS file
  - the response to be predicted/learned
  - the features extracted
  - the known perturbations
    - patient: age, treatment, etc.
    - FCS file: batch effect, labelling, etc.

Cell population analyses

- Features are extracted from cell populations that result from
  - clustering
  - manual gating

- Input data, organized as a matrix
  - For each sample/FCS file (statistical subject), we extract/collect:
    - Associated biological/medical information
    - Number of cells in each cell population
    - MFI for each functional marker



Visualization and Statistical tests

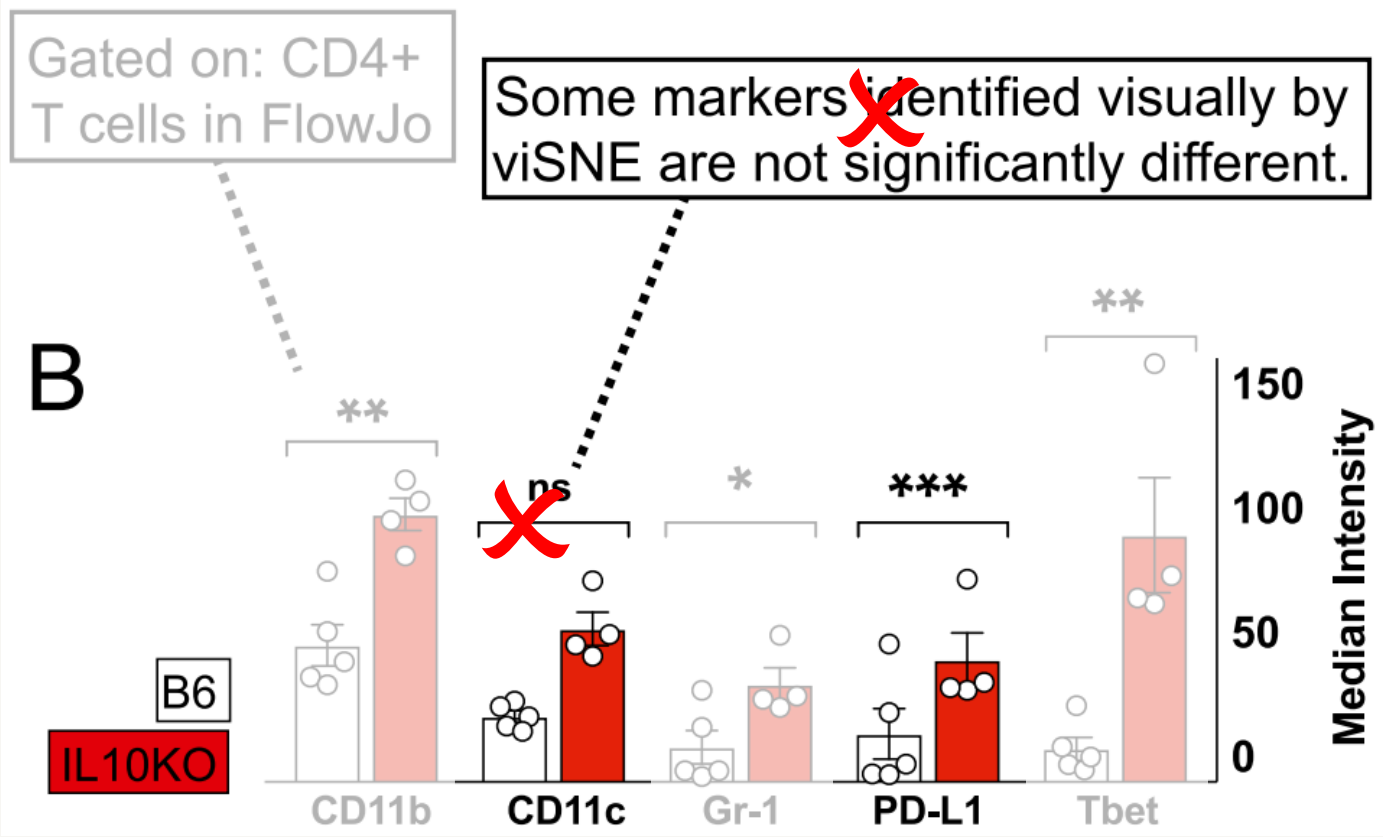
- Data transformation that I do
  - adapt range for display, to get a global view of all features at once
  - make the distribution of feature variation similar to a symmetric bell shape

- MFI
  - transformed as for bi-parametric plot: asinh, biexp, etc.

- Counts
  - percentages
  - transformation?

Testing caveats

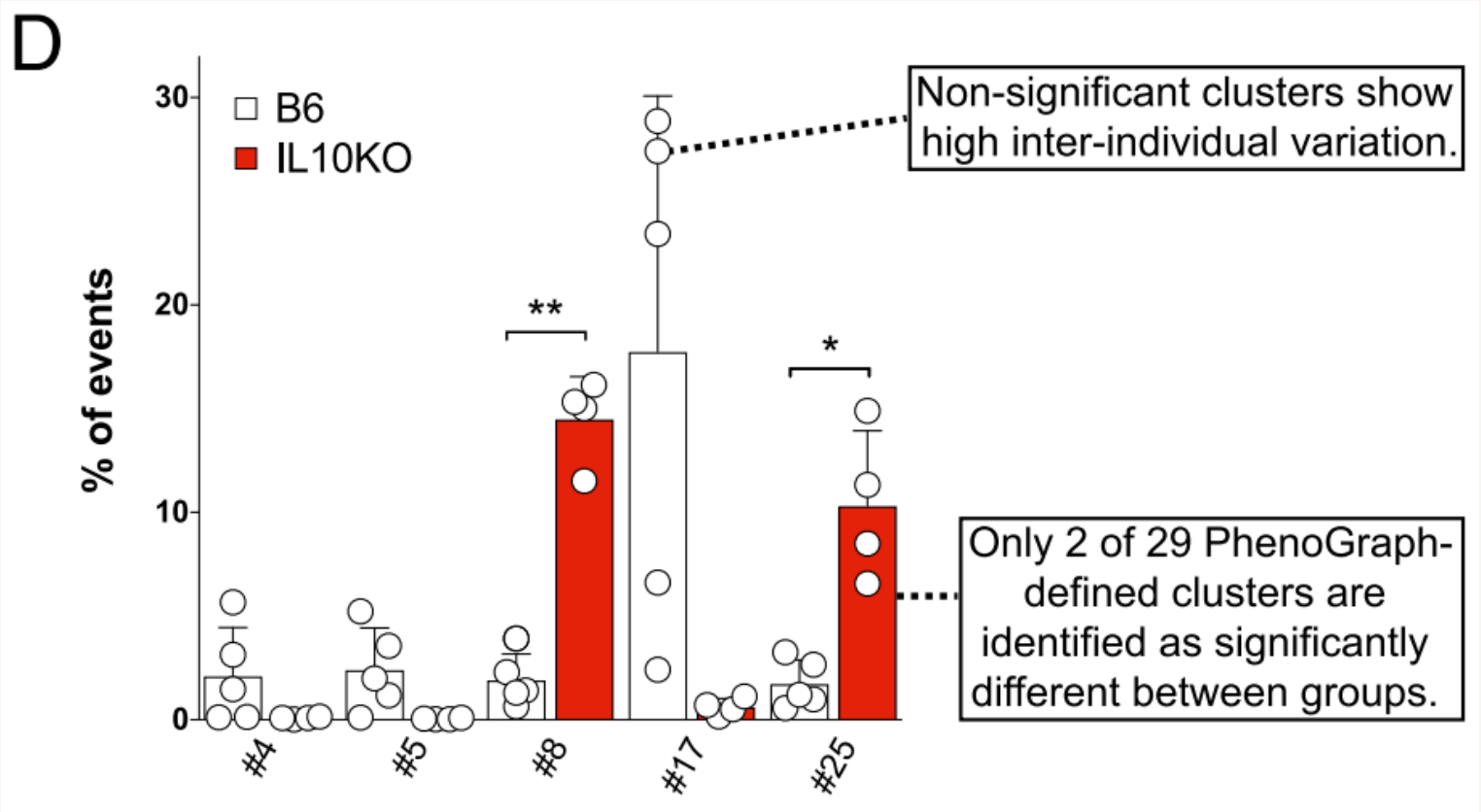
- Automate testing otherwise conclusions could be wrong



[Kimball et al., 2018]

Counts, percentages

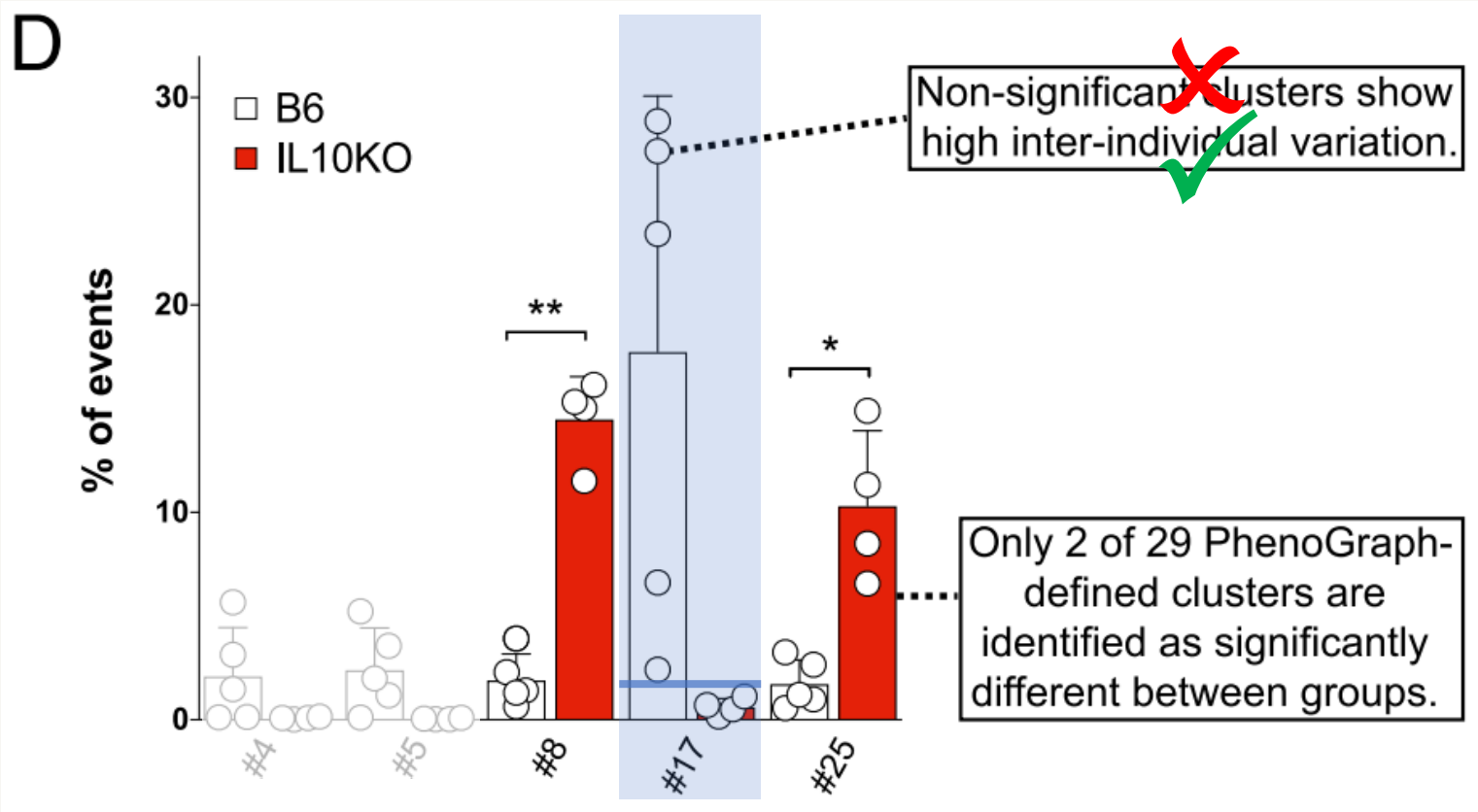
- counts depend on total
- a percentage is bounded [0..1]
- its distribution flattens out at the extremes
- can not always apply parametric tests
  - transformation
    eg asin(sqrt(p))



Kimball et al., 2018

Percentage caveats

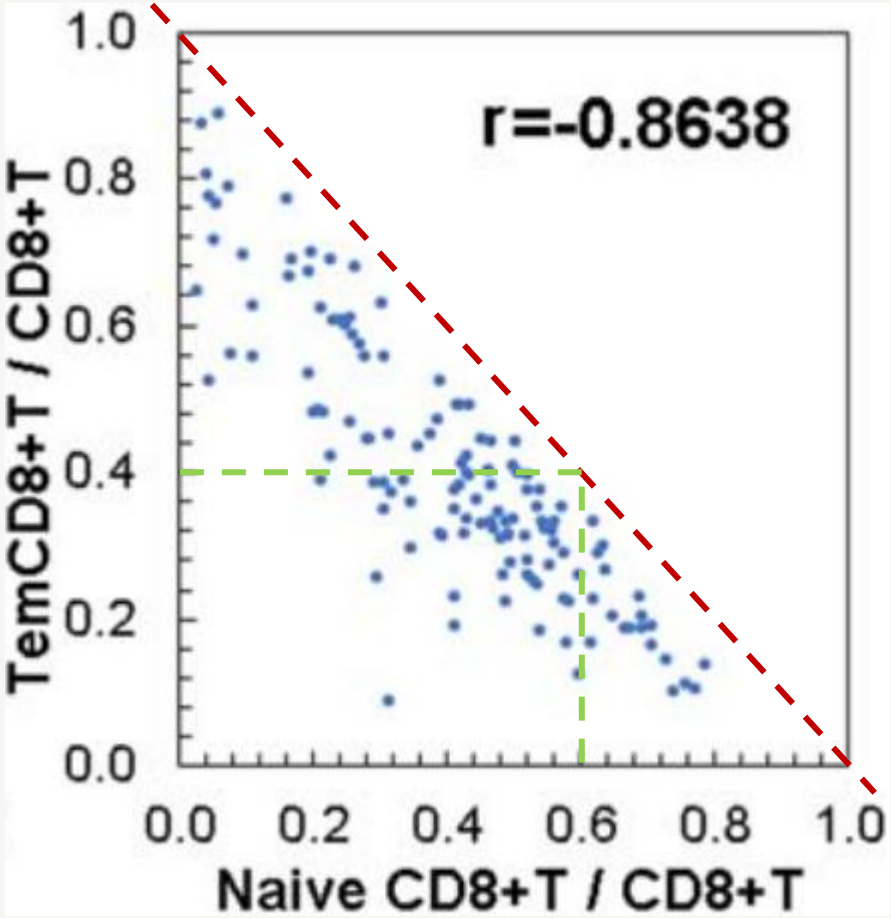
- Distribution flatten at extremes
- High variation
- Select the right test otherwise conclusions could be wrong



[Kimball et al., 2018]

Counts, percentages

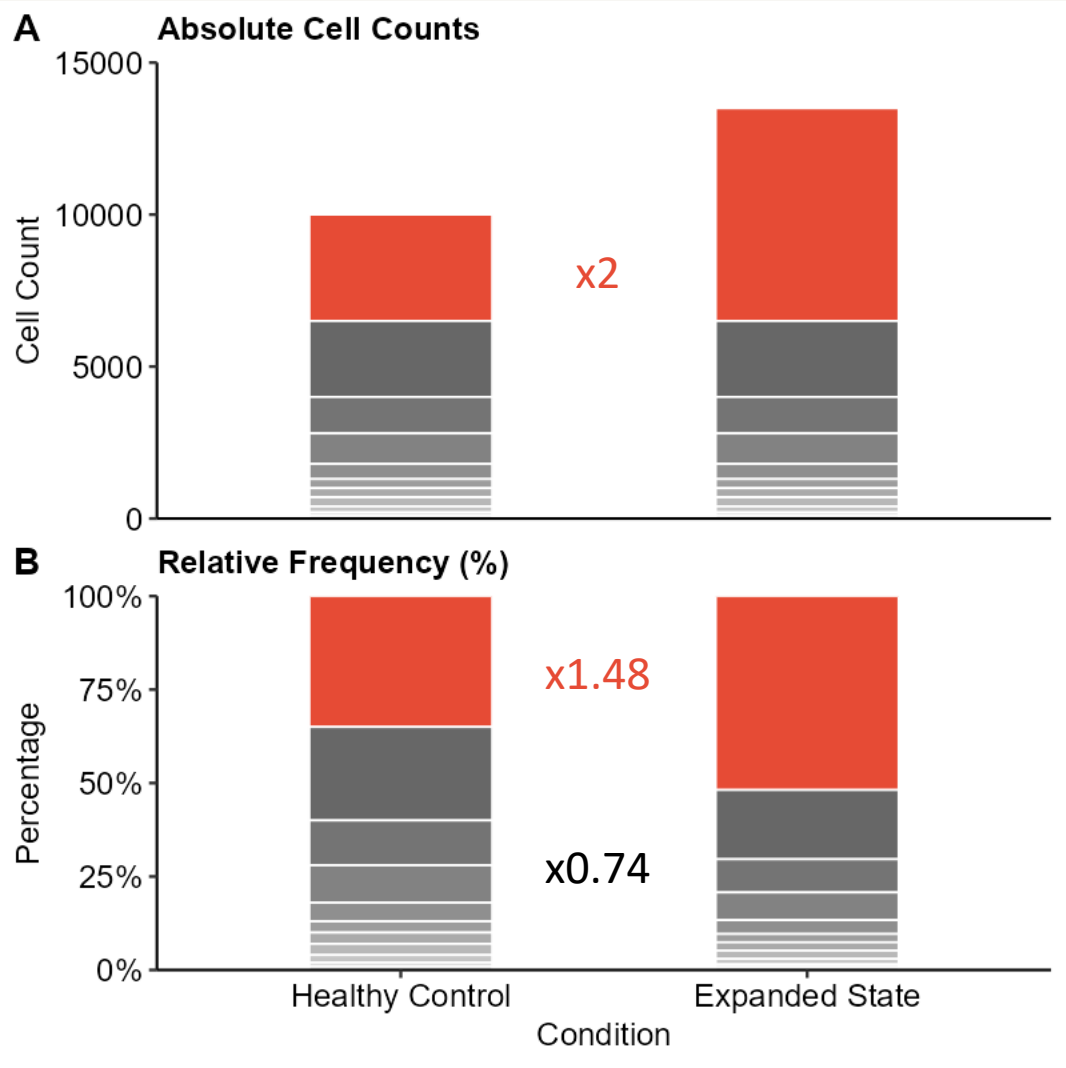
- percentage = count / reference
- what reference?
  - total count is natural/naive
  - count of a specific population? (or a specific quantity?)
  - a pool of invariant populations?
- sum of all % = 1



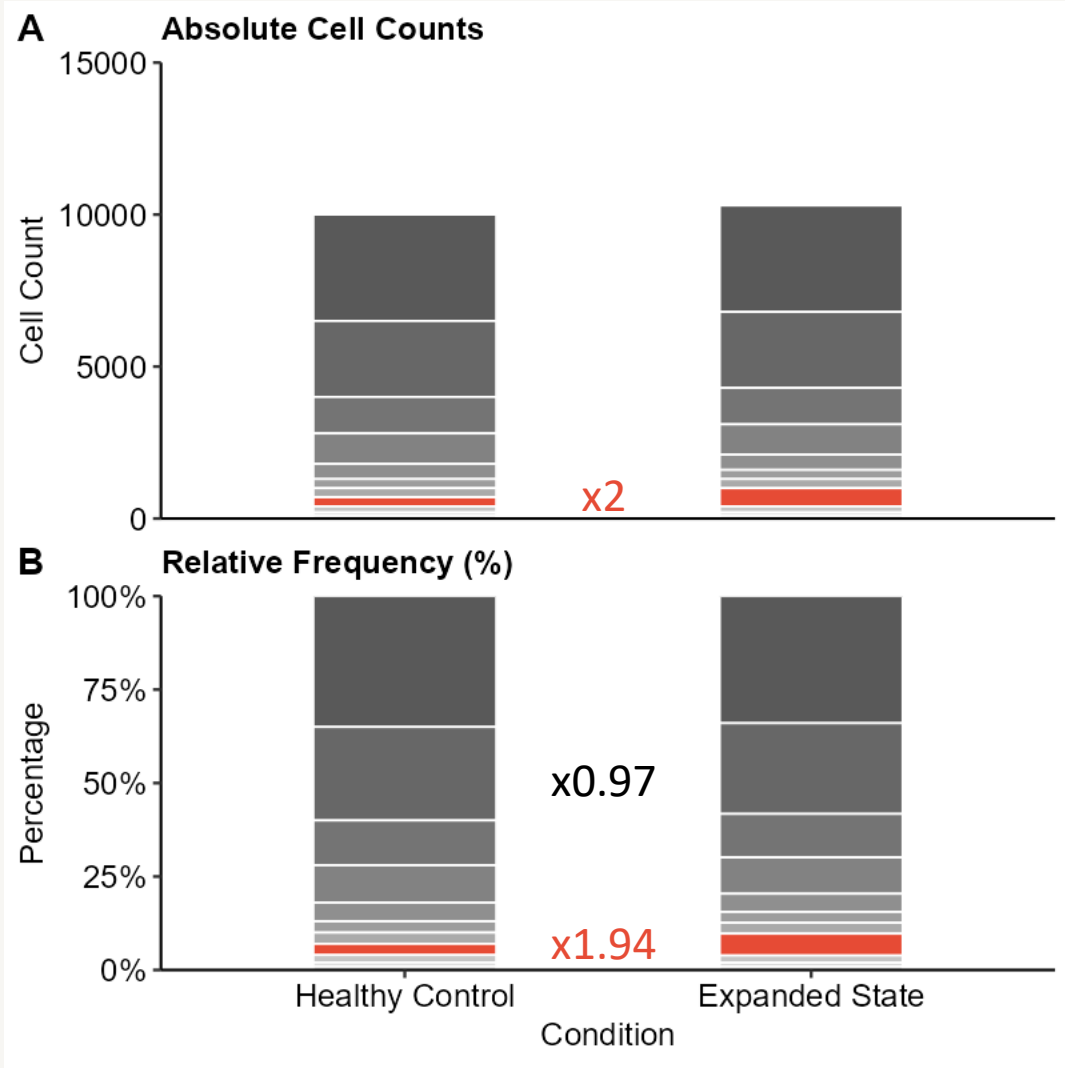
Lee et al., 2018

Counts, percentages

A large population changes



A small population changes



Compositional Data Analysis (CoDA)

- Traditional statistical methods produce spurious correlations. Compositional analysis solves this by transforming proportions into unbounded log-ratios.
- CLR: Centered Log-Ratio
  - log then mean centering (alt. median centering in DESeq2, or TMM in edgeR)
- ALR: Additive Log-Ratio
  - log then reference centering
- ILR: Isometric Log-Ratio
  - maps compositions onto an orthogonal Euclidean space

- Imputation is still required to manage zero counts (Log problem)

Aitchison, J. (1984), Pawlowsky-Glahn et al. (2006), Pawlowsky-Glahn et al. (2022)

Best Practices

- Be aware of the risks associated with traditional approaches
- Adjust p-values for multiple testing (usually FDR)
- Follow best practices for scRNA-seq (or microbiome) for counts
  - based on the negative binomial distribution, taking the value zero into account
  - however, overdispersion may still be observed

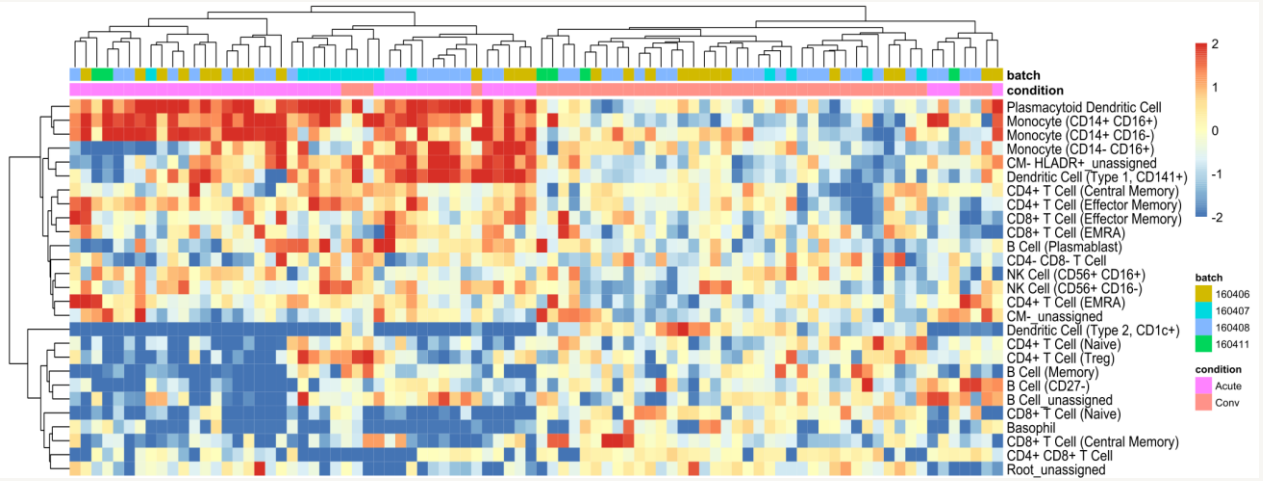
Normalize Samples and Features

- Normalize within sample and implicitly between samples



- Normalize between features
  - centering and scaling if needed (standard deviation)
  - no a priori: all samples
  - hypothesis driven: samples from 1 condition

columns = samples
rows = features



Conv condition is reference for centering, no scaling

- Details in [Lohmann et al., 2026]

ML in practice - Study

- Identify immune biomarkers predictive of the efficacy of combination immunotherapy in patients with advanced cervical cancer
- Current efficacy is 53%
- 29 patients whose samples were analyzed by mass cytometry
- 3 immune panels (T, NK, Myelo) of 40 markers
- (optSNE + Phenograph) => clusters
- 2 clusters merged in 1 candidate
- validated by manual gating
- clusters + manual gating => prepared % => 11 ML algorithms

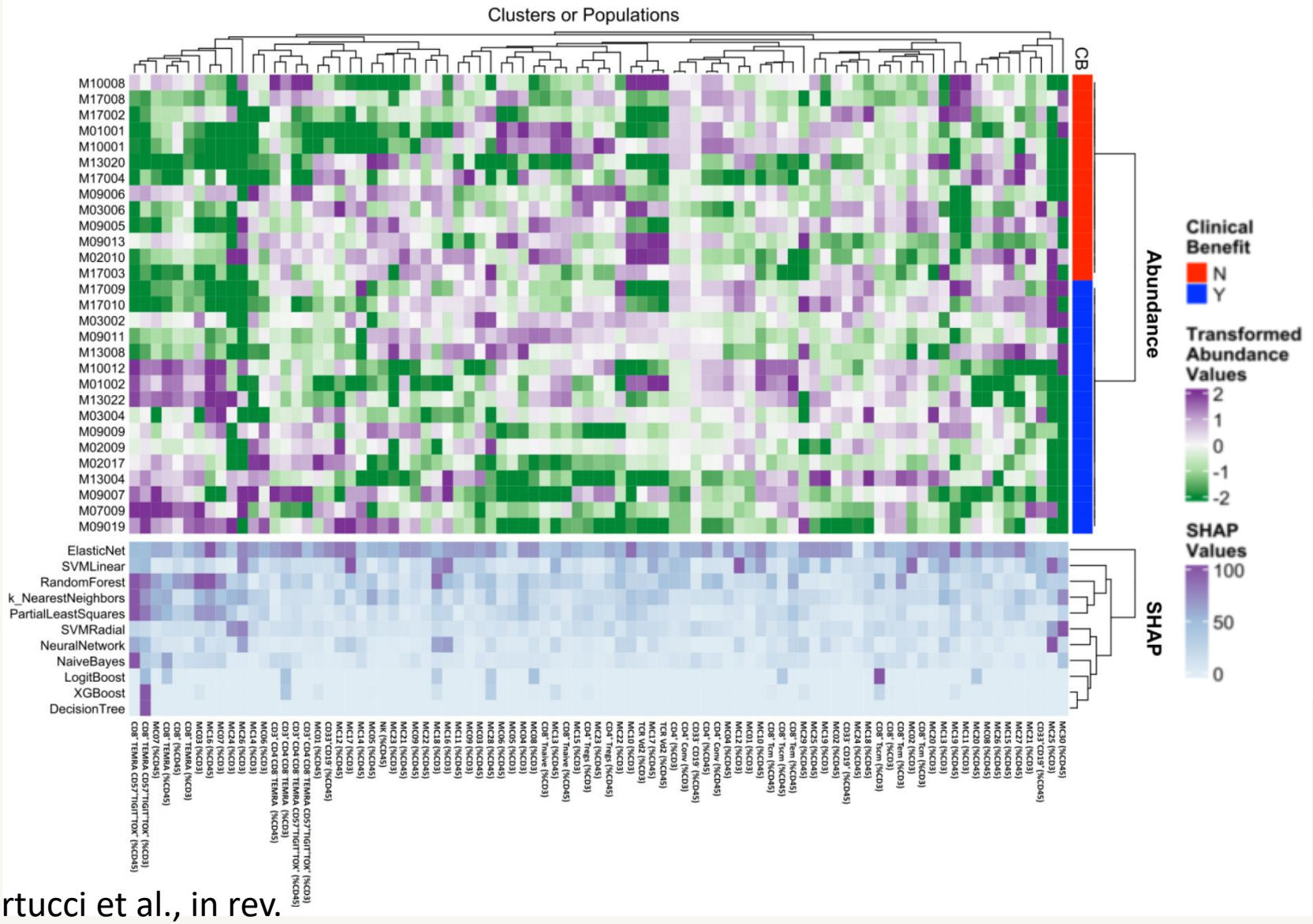
Bertucci et al., in rev.

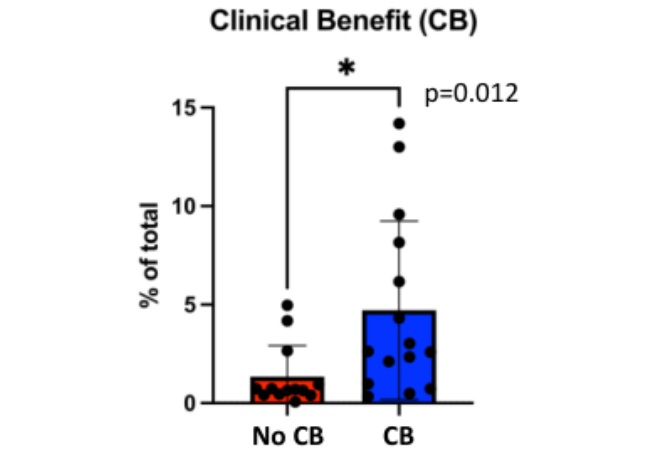
ML in practice - Data Preparation Focus

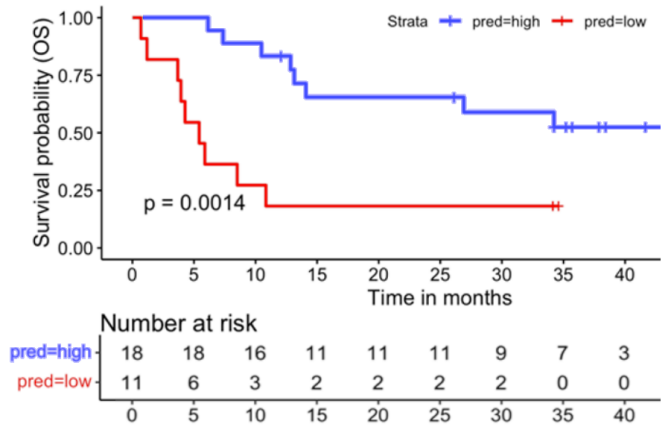
Algorithm	Scaling required	transformation benefit	noise sensitive
Elastic Net	penalty treats all coefficients equally	yes	
Partial Least Squares	large variance dominate	yes	yes
k-Nearest Neighbors	purely distance-based	+/-	yes
Naive Bayes	depends	yes	
Neural Network	essential for stable and fast training	yes	
Linear SVM	margin depends on distances	yes	
Radial SVM	kernel is based on Euclidean distance	+/-	
Logit Boost	no, ranking of values	no	
XG Boost	no, ranking of values	no	
decision tree	no, ranking of values	no	
Random Forest	no, ranking of values	no	

More in Table 1 of Greener et al., 2020

ML in practice - Results







Bertucci et al., in rev.

Take home messages

- **ML** aims to **predict** the response **using multiple features**
- **XAI** allows us to **evaluate** the **importance of features**
- The **input** dataset must be **representative**
  and include a large **enough** number of **subjects**

- Start simple, e.g. LASSO (multi-variate linear regression with penalization)
- Prepare features (e.g. LASSO) or not (e.g. RandomForest)

- Start with guidance